Brain Chemicals

While communication in the brain relies on electric pulses flashing along wirelike nerve cells, the activity of these cells—and the mental and physical states they induce—are heavily influenced by chemicals called neurotransmitters.

IS TECHNOLOGY ADDICTION THE SAME AS DRUG ADDICTION?

No, technology addiction is more comparable to overeating. Release of dopamine can increase by 75 percent when playing video games and by 350 percent when using cocaine.

Neurotransmitters

Neurotransmitters are active at the synapse, the tiny gap between the axon of one cell and a dendrite of another (see p.23). Some neurotransmitters are excitatory, meaning that they help continue the transmission of an electrical nerve impulse to the receiving dendrite. Inhibitory neurotransmitters have the opposite effect. They create an elevated negative electrical charge, which stops the transmission of the nerve impulse by preventing depolarization from taking place. Other neurotransmitters, called neuromodulators, modulate the activity of other neurons in the brain. Neuromodulators spend more time at the synapse, so they have more time to affect neurons.

TYPES OF NEUROTRANSMITTERS

There are at least 100 neurotransmitters, some of which are listed below. Whether a neurotransmitter is excitatory or inhibitory is determined by the presynaptic neuron that released it.

NEUROTRANSMITTER CHEMICAL NAME	USUAL POSTSYNAPTIC EFFECT
Acetylcholine	Mostly excitatory
Gamma-aminobutyric acid (GABA)	Inhibitory
Glutamate	Excitatory
Dopamine	Excitatory and inhibitory
Noradrenaline	Mostly excitatory
Serotonin	Inhibitory
Histamine	Excitatory

Drugs

Chemicals that change mental and physical states, both legal and illegal, generally act by interacting with a neurotransmitter. For example, caffeine blocks adenosine receptors, which has the effect of increasing wakefulness. Alcohol stimulates GABA receptors and inhibits glutamate, both inhibiting neural activity in general. Nicotine activates the receptors for acetylcholine, which has several effects, including an increase in attention as well as elevated heart rate and blood pressure. Both alcohol and nicotine have been linked to an elevation of dopamine in the brain, which is what leads to their highly addictive qualities.

TYPE OF DRUG	EFFECTS
Agonist	A brain chemical that stimulates the receptor associated with a particular neurotransmitter, elevating its effects.
Antagonist	A molecule that does the opposite of an agonist, by inhibiting the action of receptors associated with a neurotransmitter.
Reuptake inhibitor	A chemical that stops a neurotransmitter from being reabsorbed by the sending neuron, thus causing an agonistic response.

BLACK WIDOW SPIDER VENOM INCREASES LEVELS OF THE **NEUROTRANSMITTER ACETYLCHOLINE,** WHICH CAUSES **MUSCLE SPASMS**